Isosceles and equilateral triangles

ANSWER KEY

Core Concepts: Isosceles and equilateral triangles

1 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 7

c. 5 ✓

d. 7

Substituting $x = 1$ into $y = 2x + 3$ gives $y = 2(1) + 3 = 5$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 5

b. 9

c. 7 ✓

d. 9

Substituting $x = 2$ into $y = 2x + 3$ gives $y = 2(2) + 3 = 7$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 59^\circ) = 88^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 7

b. 11

c. 9 ✓

d. 11

Substituting $x = 3$ into $y = 2x + 3$ gives $y = 2(3) + 3 = 9$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 66^\circ) = 81^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 9

b. 13

c. 11 ✓

d. 13

Substituting $x = 4$ into $y = 2x + 3$ gives $y = 2(4) + 3 = 11$.

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